

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
import warnings
warnings.filterwarnings('ignore')

df= pd.read_csv('IPL.csv')
df.head()
```

Next steps:

[Generate code with df](#)[New interactive sheet](#)

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 74 entries, 0 to 73
Data columns (total 20 columns):
#   Column                Non-Null Count  Dtype
---  -
0   match_id              74 non-null    int64
1   date                  74 non-null    object
2   venue                 74 non-null    object
3   team1                 74 non-null    object
4   team2                 74 non-null    object
5   stage                 74 non-null    object
6   toss_winner           74 non-null    object
7   toss_decision         74 non-null    object
8   first_ings_score      74 non-null    int64
9   first_ings_wkts       74 non-null    int64
10  second_ings_score     74 non-null    int64
```

```
11 second_ings_wkts      74 non-null    int64
12 match_winner          74 non-null    object
13 won_by                 74 non-null    object
14 margin                 74 non-null    int64
15 player_of_the_match    74 non-null    object
16 top_scorer             74 non-null    object
17 highscore              74 non-null    int64
18 best_bowling           74 non-null    object
19 best_bowling_figure    74 non-null    object
dtypes: int64(7), object(13)
memory usage: 11.7+ KB
```

```
print(f"your rows are {df.shape[0]}, and columns are {df.shape[1]}")
```

```
your rows are 74, and columns are 20
```

```
df.isnull().sum()
```

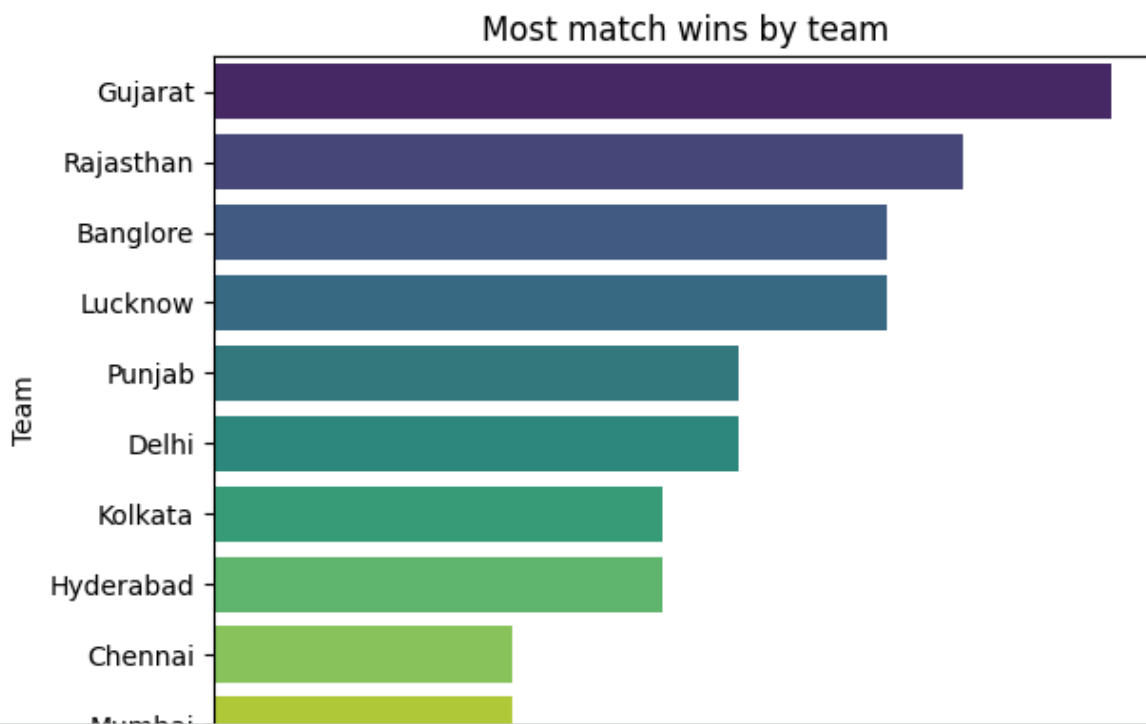
	0
match_id	0
date	0
venue	0
team1	0
team2	0
stage	0
toss_winner	0
toss_decision	0
first_ings_score	0
first_ings_wkts	0
second_ings_score	0
second_ings_wkts	0
match_winner	0
won_by	0
margin	0
player_of_the_match	0
top_scorer	0
highscore	0
best_bowling	0
best_bowling_figure	0

```
dtype: int64
```

Which team won the most matches?

```
match_wins = df['match_winner'].value_counts()
sns.barplot(y= match_wins.index, x= match_wins.values, palette='viridis')
plt.title('Most match wins by team')
plt.xlabel('Wins')
plt.ylabel('Team')
```

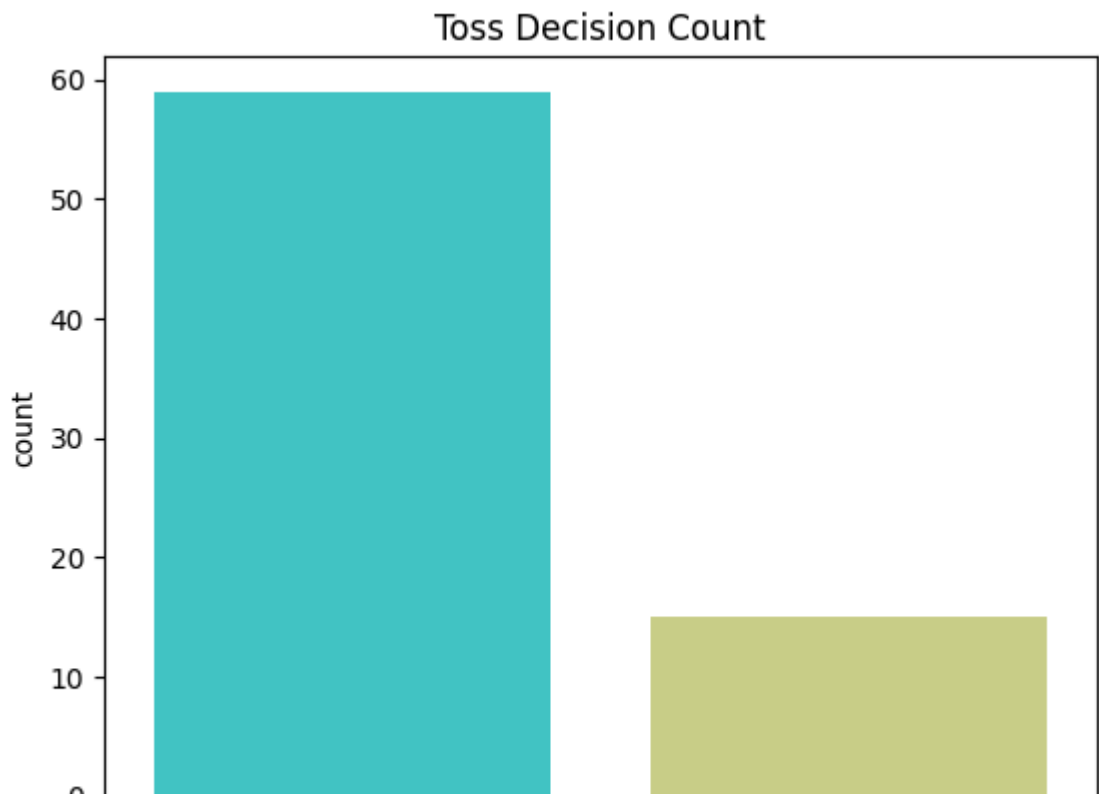
```
Text(0, 0.5, 'Team')
```



TOSS DECISION

```
sns.countplot(x = df['toss_decision'],palette='rainbow')
plt.title('Toss Decision Count')
```

```
Text(0.5, 1.0, 'Toss Decision Count')
```



Toss Winner vs Match Winner

```
count = df[df['toss_winner'] == df['match_winner']]['match_id'].count()  
count
```

```
np.int64(36)
```

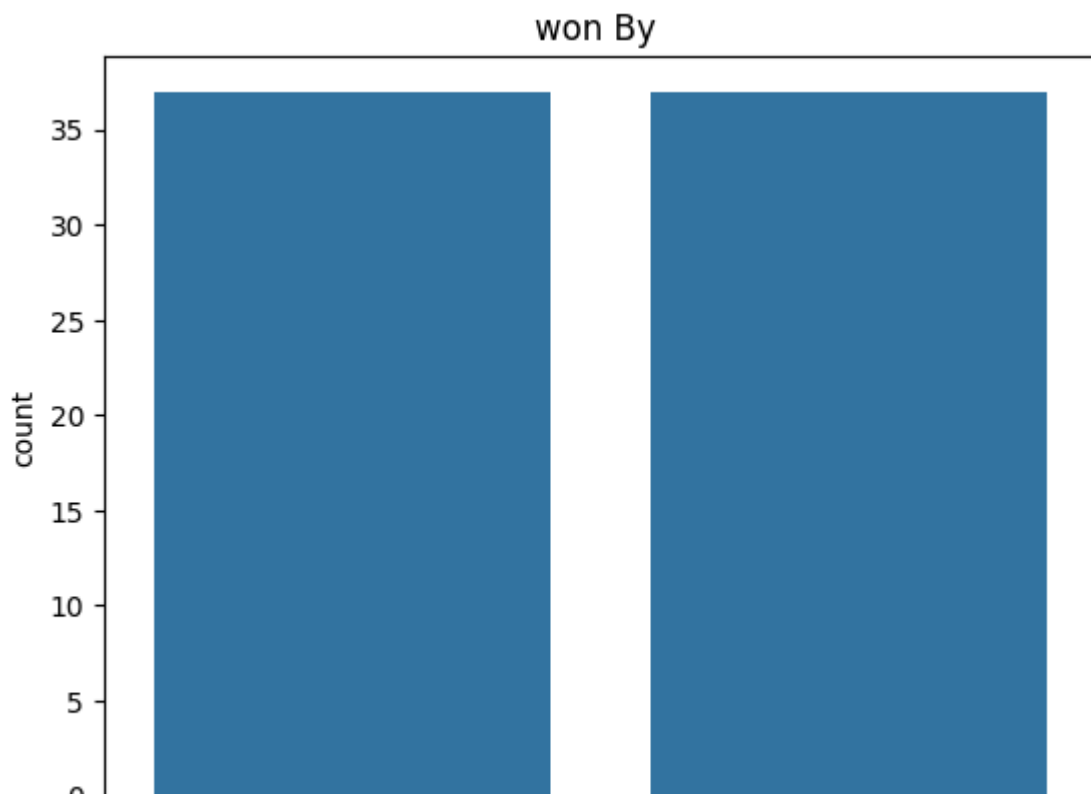
```
count = df[df['toss_winner'] == df['match_winner']]['match_id'].count()  
percentage = (count*100)/df.shape[0]  
percentage.round(2)
```

```
np.float64(48.65)
```

How do teams win? (Runs vs Wickets)

```
sns.countplot(x = df['won_by'])  
plt.title("won By ")
```

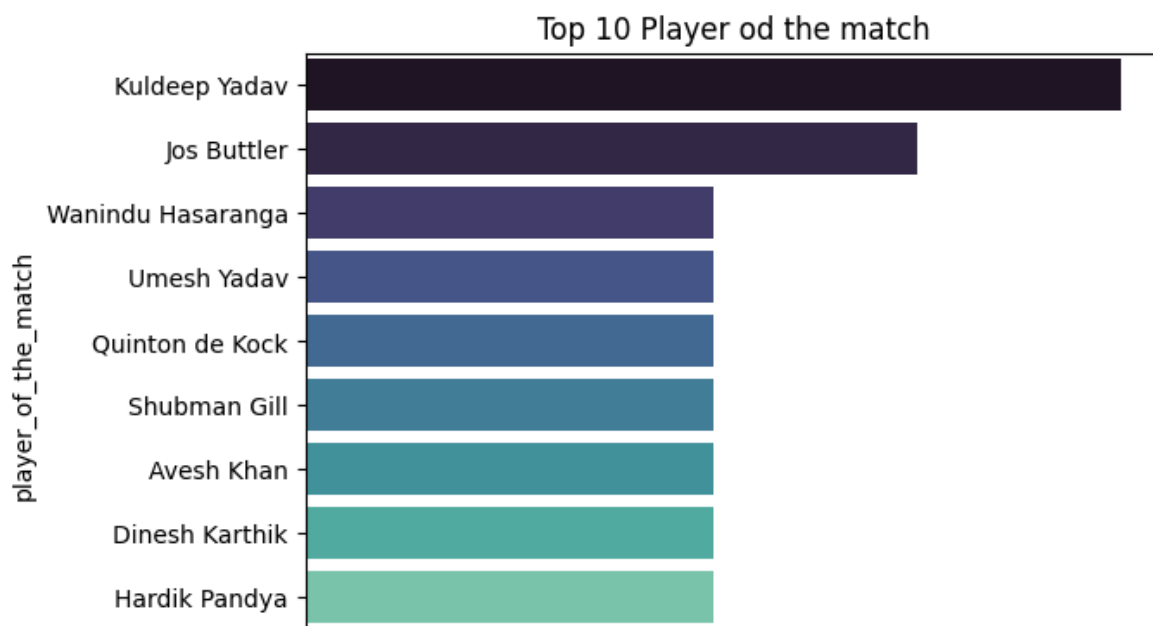
```
Text(0.5, 1.0, 'won By ')
```



Key Player Performances

```
count = df['player_of_the_match'].value_counts().head(10)
sns.barplot(x = count.values,y = count.index,palette='mako')
plt.title("Top 10 Player od the match")
```

```
Text(0.5, 1.0, 'Top 10 Player od the match')
```



Top Scorers

```
df['top_scorer'].head(5)
```

	top_scorer
0	MS Dhoni
1	Ishan Kishan
2	Faf du Plessis
3	Deepak Hooda
4	Aiden Markram

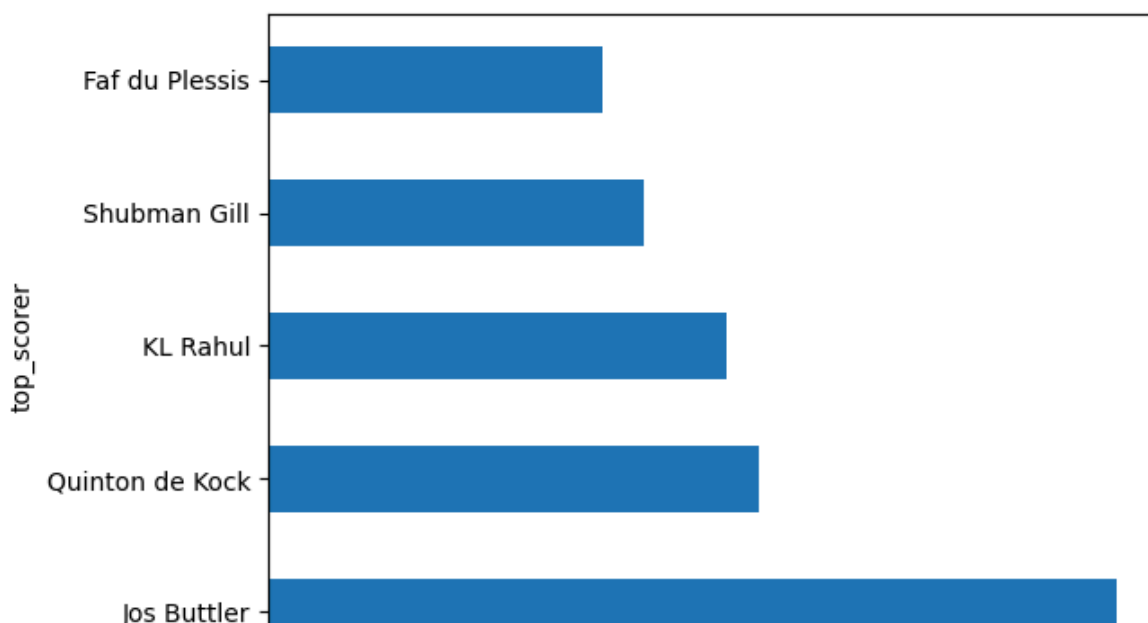
dtype: object

```
high = df.groupby('top_scorer')['highscore'].sum()  
high
```


highscore	
top_scorer	
Abhishek Sharma	122
Aiden Markram	57
Andre Russell	119
David Miller	94
David Warner	213

```
high = df.groupby('top_scorer')['highscore'].sum().sort_values(ascending=False)
high.plot(kind = 'barh')
```

```
<Axes: <Figure> by top_scorer: 66
```



Nitish Rana	57
Best Bowling Figures	
Pat Cummins	56

```
df.head()
```

Rahul Tripathi	147
Rajat Patidar	112
Riyan Parag	56
Ruturaj Gaikwad	99
Sai Sudharsan	65
Sanju Samson	54
Sherfane Rutherford	28
Shikhar Dhawan	158
Shimron Hetmyer	59
Shivam Dube	95

	Shubman Gill		288					
match_id	date	venue	team1	team2	stage	toss_winner	t	
0	1	March 26, 2022	Wankhede Stadium, Mumbai	Chennai	Kolkata	Group	Kolkata	
1	2	March 27, 2022	Brabourne Stadium, Mumbai	Delhi	Mumbai	Group	Delhi	
2	3	March 27, 2022	Dr DY Patil Sports Academy, Mumbai	Banglore	Punjab	Group	Punjab	
3	4	March 28, 2022	Wankhede Stadium, Mumbai	Gujarat	Lucknow	Group	Gujarat	
4	5	March 29, 2022	Maharashtra Cricket Association Stadium, Pune	Hyderabad	Rajasthan	Group	Hyderabad	

Next steps:

[Generate code with df](#)[New interactive sheet](#)

df['best_bowling_figure'].head(5)

	best_bowling_figure
0	3--20
1	3--18
2	2--59
3	3--25
4	3--22

dtype: object

```
df['best_bowling_figure'] = df['best_bowling_figure'].apply(lambda x : x.split('-'))
df['best_bowling_figure'] = df['best_bowling_figure'].astype(int)
top_bowlers = df.groupby('best_bowling')['best_bowling_figure'].sum().sort_values(ascending=False)
top_bowlers_df = top_bowlers.reset_index()
top_bowlers_df.columns = ['bowler', 'total_best_figures']
```

```
df, y='bowler', x='total_best_figures', palette='viridis')
```

<Axes: xlabel='total_best_figures', ylabel='bowler'>



Most Matches Played by Venue

Uppan Malik
Wanindu Hasaranga

```
venue_count = df['venue'].value_counts()  
sns.barplot(y = venue_count.index, x = venue_count.values, palette='magma')
```

<Axes: ylabel='venue'>

